

Question	Answer	Marks
6(a)(i)	Binding energy is defined as the amount of energy required to completely separate a nucleus into its individual protons and neutrons or nucleons.	1
6(a)(ii)	At $A=170$, $BE/A \sim 8.10$ MeV (between 8.10 to 8.30 MeV) BE of ${}^{170}_{77}\text{Ir} = (8.10 \text{ to } 8.30) \times 170 = 1377 \text{ MeV to } 1411 \text{ MeV}$	1
6(a)(iii)	$B.E = \Delta mc^2$ $\Delta m = \frac{BE}{c^2} = \frac{1377 \times 10^6 \times 1.6 \times 10^{-19}}{(1.66 \times 10^{-27})(3 \times 10^8)^2} = 1.475 \text{ u}$ 1377 MeV to 1411 MeV $\rightarrow$ 1.475 u to 1.515 u	1 1
6(b)(i)	Mass Difference = $(3 \times 4.002603 - 12.000000) \text{ u}$ Energy released $E = \Delta mc^2$ $= 0.007809 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 1.17 \times 10^{-12} \text{ J}$	1 1
6(b)(ii)	After this point, after nickel which has the highest binding energy per nucleon , the <u>total binding energy of the reactants exceeds that of the products</u> , so no net energy would be released after the reaction.	1

	Max Marks	7
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Question	Answer	Marks
7(a)(i)	To prevent microwaves/(em) radiation from leaking/escaping out/exiting of the	1